Microwave Technology for Brain Activities Detection of Rats

Xing Jiang, Bo Kang, Xiaoming Li, Lin Peng
Guangxi Wireless Broadband Communication and Signal Processing Key Laboratory
Key Laboratory of Microwave and Optical Wave-applied Technology
Guilin University of Electronic Technology, Guilin, 541004, Guangxi, China
E-mail:1003439204@qq.com

***Abstract*—A new detection approach based on microwave transmission approach for brain neuronal activities detection is presented in this paper. The existing research shows that the dielectric at brain excited site has a dynamic nature, which gives a hope for brain neuronal activities detection by using microwave approach. Here, we discuss the principles of microwave detecting method for brain neuronal activities. A simplified model was applied in simulation to study the electromagnetic energy distribution and poynting vector. To proof the approach, we have performed an intensive study of the phase change when an EM wave propagating through the function site such that the permittivity varies. The results demonstrates that microwave can be used to detect brain activities.**

Keywords—Microwave detection technology; brain neuronal activities; electromagnetic field distribution in rat head; power flow; phase change

I. INTRODUCTION

Over the past decades, a lot of researches have been conducted on the application of microwave in the biological medicine. The advantages of using microwave technology are non-ionization, non-contact, low cost implementation, easy operation and user friendliness. Which are of many benefits in monitoring physiological information, diagnosing diseases, and providing therapies for patients. Especially, there has been a tremendous interest in using microwave techniques for the detection of abnormal tissue, such as breast cancer detection and brain stroke diagnosis. Due to the difference of detectable in the dielectric between the abnormal tissue and the surrounding normal issue at microwave frequencies. A microwave system has been presented for head imaging to detect brain injuries[1]. It was successful to acquire the tumor location and differentiate hemorrhagic from ischemic stroke in acute stroke patients[2-4]. Also, microwave based radiometry has been studied for non-invasive brain temperature monitoring[5].

The brain has complex functions and particular structures, it always becomes the hot-spot and attracted numerous researchers to be engaged in demystifying the most sophisticated organ. The existing technologies for detect neuronal signal include scalp electroencephalography (EEG), functional magnetic resonant imaging (fMRI), positron emission tomography (PET) and so on. Which have been successfully and broadly applied to various brain activity studies. Recently, based on the changes of dielectric properties of cerebral spinal fluid, the possibility of using microwave for detecting of Alzheimer's disease has been studied. It gives a hope to detect Alzheimer's disease up to a decade before the onset of symptoms[6].

The dielectric varies with the ion concentration of the extra-cellular fluid surrounding neurons in activation at a brain functional site. So the phase of the electromagnetic wave varies with the dynamic dielectric. In this paper, we aimed to analyze the electromagnetic field distribution and poynting vector within the rat head and extract brain activities of rat from the phase variation of transmission coefficient. The proposed method of microwave transmission approach for brain activities detection has its advantages on non-contact and low-cost properties in contrast to scalp EEG. The purpose for researching of rats' neuronal activities detection will provide a guidance for human neuronal activities detection in future.

II. PRINCIPLES OF MEASUREMENT

The theory of the proposed detection approach is based on analysis of EM signals transmitted through the brain. We know that certain brain activities are always closely related to certain brain functional site. In specific activated site, the ion concentration of the extra-cellular fluid that surrounded by the neurons of the brain is varying with the neuronal activation[7]. Meanwhile, the extra-cellular fluid surrounding neurons appear to be a dynamic dielectric.

When microwave travels through the dynamic dielectric, the characteristics of amplitude and phase will be changed with the variation of electric properties of activated area. Therefore, the neuronal activity of functional site can be sensed by the phase change of transmission coefficient-S_{21}. A three-layer simplified model of a single neuron for functional site is shown in Fig.1. When the microwave pass through a mass of neurons, the phase variation $\Delta\Psi$ between transmitting and receiving antennas transmission coefficient is given by

$$\Delta \psi = \sum_{i=1}^{n} \gamma_i(\varepsilon_i, \sigma_i) d_i + \sum_{j=1}^{m} Arg\,[\tau_j(\varepsilon_i, \sigma_i)] \qquad (1)$$

Where γ_i is the propagation constant of the ith layer, given by $j\omega\sqrt{\mu\varepsilon_i}$, ε_i and σ_i are the dielectric constant and conductivity of

978-1-5090-4743-7/16/$31.00 © 2016 IEEE

the ith layer, respectively . d_i is the propagating path of the ith layer, and $Arg(\tau_j)$ is the phase of the propagation coefficient of the interface.

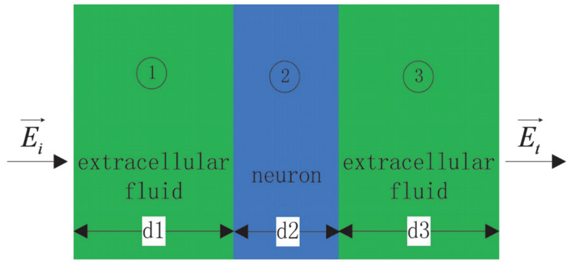


Fig 1. Simplified concept model of a single neuron for functional site

According Equation (1), although the fact that total ion numbers of the functional site are not changed during the process of neuronal activities, the phase of transmission coefficients keeps changing with the neuronal activities. Hence, microwave technology has great potential for neuronal activities detection of rats.

III. SIMULATION ANALYSIS

Biological tissue is a lossy transmission medium, which presents several challenges by itself. Especially, it's difficult for electromagnetic wave to involve in brain to acquire neuronal activities, due to the reflection of the incident waves from the interface of different tissue. The multi-path caused by the boundaries between tissues would lead the dispersed wave to be completely untraceable at the receiver.

A. *Simulation analysis of electromagnetic fields distribution*

To investigate the electromagnetic fields distribution within the rat brain, a simplified rat head phantom with functional site and EM simulation software CST2015 has been used to analyze the electrical properties. The simulation model is illustrated in Fig. 2. The dielectric properties of tissues were introduced in [8][9]. The relative permittivity of functional site and head tissue is 17.2 and 11.4 at 30 GHz, respectively. The head size of rat is approximately 12 mm from the front to the back, 16mm from the left to the right, and 10 mm from the top to the bottom. Standard waveguide operating at 30 GHz with the dimension of 7.11 mm × 3.56 mm is adopted for our simulation, and operate on the principle of transverse electric. In order to let more electromagnetic wave through the rat head to arrive at the receiver, the transmitting waveguide was positioned close to the rat head.

The distribution of electromagnetic field in rat head is presented in Fig. 3. Only the most of interesting and useful field plots are displayed. It can be noticed that a strong reflection of the incident waves from the air-skin interface occurs causing half of the energy to be back-scattered. The free space between the antenna and rat head also introduces additional attenuation. Moreover, due to the presence of the surface waves the energy penetrating into the functional site is lower than -30dB. However, appreciable intensity of electromagnetic energy can be detected in the functional site. It demonstrates potential capability of microwave technology to detect brain activities.

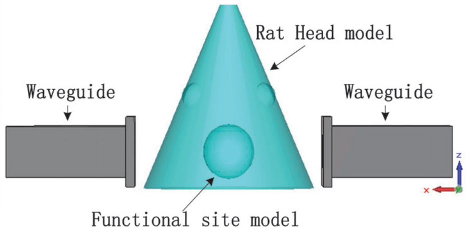
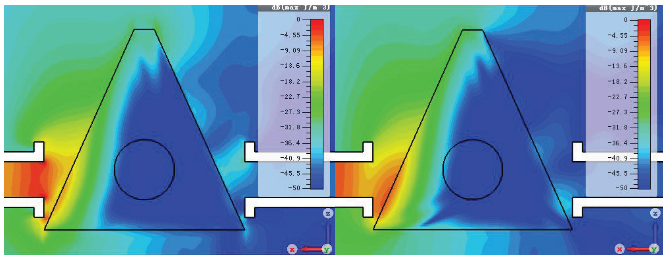


Fig 2. Simulation phantom.



(a) (b)

Fig. 3. Electromagnetic energy density within phantom. (a) electric energy density in xz plane. (b) magnetic energy density in xz plane.

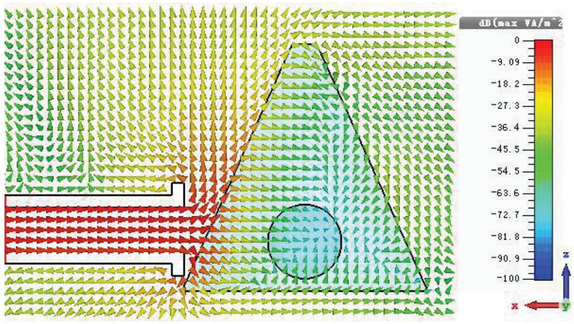


Fig. 4. Magnitude and direction of poynting vector for selected waveguide positions at 30GHz.

B. *Simulation analysis of power flow*

To investigate the direction of power flow, the normalized poynting vector ($\boldsymbol{S}=\boldsymbol{E}\times\boldsymbol{H}^*$) is plotted on a logarithmic scale, as shown in Fig. 4. Only -30dB of the energy reaches the receiver, however, it is still in the range of the detectable accuracy of vector network analyzer. According to the path of power flow, the optimal location for the transceiver is suggested to be located in radiating near-field region.

C. *Simulation analysis of time-varying functional site*

To confirm the analysis, we used EM simulation software-FEKO to simulation the phase change of a EM wave propagating through activated area with time-varying permittivity. The material properties of head tissue are shown in Table 1. where f is the changed frequency of dielectric in excited areas, the value is 3 Hz. t is the simulation recording time.

The variation of the phase of EM wave propagating through the rat brain was analyzed using FFT. The results are shown in Fig. 5. The first parts of (a) shows the phase of the EM wave changed from 66.4651 to 66.4663 degrees, and the second parts of (b) shows the spectral densities of the phase change of transmission coefficient. The dominant variation frequency of phase is still 3 Hz, which consistent with the change frequency of dielectric constant. The simulation results prove that neuronal activity of brain functional site can be measured by phase change of transmission coefficient-S_{21} when the microwave pass through activated area. An experiment based microwave technology of rats' brain activities detection had been conducted cooperate with Neuroengineering Lab of National University of Singapore. And the results of tests show that Delta Wave of rats can be detected by microwave[8].

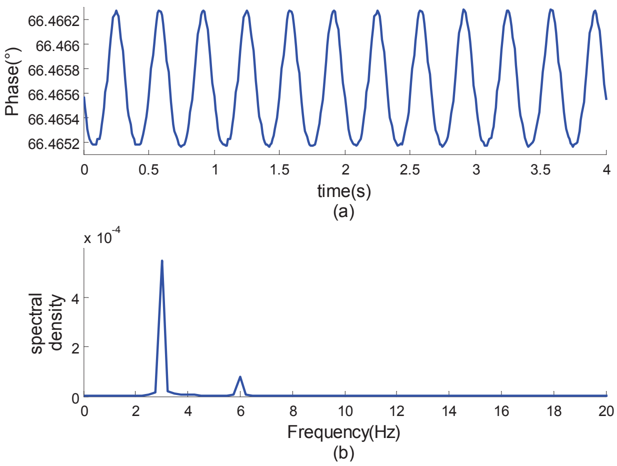


Fig 5. Simulation results. (a) variation of the phase change. (b) the spectral density distribution

Table 1. THE MATERIAL PROPERTIES OF THE SIMULATION MODEL

Tissue	*Electrical conductivity*	*Relative permittivity*
Functional site	27.2	$17.2+\sin(2\times pi \times f \times t)$
Brain tissue	16.0	11.4

IV. CONCLUSION

In this paper, we proposed a new method for brain neuronal activities detection based on microwave technology. The studying of electromagnetic energy density and power flow demonstrate the possibilities for non-invasive detection of brain neuronal activities by microwave. The results show that the phase of transmission coefficient changes with the dynamic permittivity in activated area. Therefore, the brain activity can be detected by phase change of transmission coefficient. In the foreseeable future, microwave detecting technology will be gradually used to detect the specific brain activity.

ACKNOWLEDGMENT

The authors would like to thank the financial supports of National Nature Science Foundation of China (Grant NOs. 61371056 & 61401110) and Innovation Project of GUET Graduate Education (YJCXS201515).

[1] B.J. Mohammed, A.M. Abbosh, and S. Mustafa, et al, "Microwave system for head imaging," IEEE Transactions on Instrumentation and Measurement, vol. 63, No.1, pp. 117-123, January 2014.

[2] M. Persson, A. Fhager, H.D. Trefna, et al, "Microwave-based stroke diagnosis making global prehospital thrombolytic treatment possible," IEEE Transactions on Biomedical Engineering, vol. 61, No.11, pp. 2806-2817, 2014.

[3] H.Y. Mesri, "Localization of hemorrhage site in stroke patients using multichannel microwave measurements," Annual International Conference of the IEEE Engineering in Medicine and Biology Society,pp. 5927-5930, August 2012.

[4] M. Klemm, I.J. Craddock, J. Leendertz, et al, "Breast cancer detection using symmetrical antenna array," The Second European Conference on Antennas and Propagation, pp.1-5, November 2007.

[5] Stauffer, P. R. et al. Non-invasive measurement of brain temperature with microwave radiometry: demonstration in a head phantom and clinical case. Neuroradiol J 27, 3–12 (2014).

[6] G.M. Noetscher, A.T. Htet, J.M. Elloian, et al, "Detecting in vivo changes of electrical properties of Cerebral Spinal Fluid using microwave signals from small coil antennas-numerical simulation," Signal Processing in Medicine and Biology Symposium, pp. 1-6, December 2012.

[7] A. Takeda, S.Imano, H. Itoh, et al, "Zinc release in the lateral nucleus of the amygdala by stimulation of the entorhinal cortex," Brain research, vol. 1118, No.1, pp. 52-57, 2006.

[8] X.P. Li, Q. Xia, D.Qu, et al, "The Dynamic Dielectric at a Brain Functional Site and an EM Wave Approach to Functional Brain Imaging," Scientific reports, November 2014.

[9] IFAC. "Dielectric Properties of Body Tissues"[Online]. Available: http://niremf. ifac.cnr.it/tissprop/.